

Intro To Computers and Python

Lec-3-4

Libraries

- The Python Standard Library provides rich capabilities for text/binary data processing, mathematics, functional-style programming, file/directory access, data persistence, data compression/archiving, cryptography, operating-system services, concurrent programming, interprocess communication, networking protocols, JSON/XML/other Internet data formats, multimedia, internationalization, GUI, debugging, profiling and more.
- Python has an enormous and rapidly growing community of open-source developers in many fields. One of the biggest reasons for Python's popularity is the extraordinary range of open-source libraries developed by the open-source community.

Other Programming Languages

- Basic was developed in the 1960s at Dartmouth College to familiarize novices with programming techniques. Many of its latest versions are object-oriented.
- C was developed in the early 1970s by Dennis Ritchie at Bell Laboratories. It initially became widely known as the UNIX operating system's development language.
- C++, which is based on C, was developed by Bjarne Stroustrup in the early 1980s at Bell Laboratories. C++ provides features that enhance the C language and adds capabilities for object-oriented programming.
- Sun Microsystems in 1991 funded an internal corporate research project led by James Gosling, which resulted in the C++-based object-oriented programming language called Java.
- C# was developed to integrate the web into computer applications and is now widely used to develop many types of applications.

Ipython Modes

- Can be run in 2 modes
- In interactive mode, you'll enter small bits of Python code called snippets and immediately see their results.
- In script mode, you'll execute code loaded from a file that has the .py extension (short for Python). Such files are called scripts or programs, and they're generally longer than the code snippets you'll do in interactive mode.

Internet and World Wide Web

- The protocol (set of rules) for communicating over the ARPANET became known as the Transmission Control Protocol (TCP). TCP ensured that messages, consisting of sequentially numbered pieces called packets, were properly delivered from sender to receiver, arrived intact and were assembled in the correct order.
- The World Wide Web (simply called “the web”) is a collection of hardware and software associated with the Internet that allows computer users to locate and view documents (with various combinations of text, graphics, animations, audios and videos) on almost any subject.
- A service that provides access to itself over the Internet is known as a web service.
- The Internet is no longer just a network of computers—it’s an Internet of Things (IoT). A thing is any object with an IP address and the ability to send, and in some cases receive, data automatically over the Internet.
- crowdsourced data—that is, the data that’s continuously supplied by their users and their users’ devices worldwide.

Software Terminologies

- Define the terms, refactoring, design patterns, cloud computing and software development kits (SDKs).
- Refactoring: Reworking programs to make them clearer and easier to maintain while preserving their correctness and functionality.
- Proven architectures for constructing flexible and maintainable object-oriented software. The field of design patterns tries to enumerate those recurring patterns, encouraging software designers to reuse them to develop better-quality software using less time, money and effort.
- Cloud computing: You can use software and data stored in the “cloud”—i.e., accessed on remote computers (or servers) via the Internet and available on demand—rather than having it stored locally on your desktop, notebook computer or mobile device.
- Software Development Kits (SDKs)—The tools and documentation that developers use to program applications.

Big Data

- What are the 4V's of Big Data?
- Volume—the amount of data the world is producing is growing exponentially.
- Velocity—the speed at which that data is being produced, the speed at which it moves through organizations and the speed at which data changes are growing quickly.
- Variety—data used to be alphanumeric (that is, consisting of alphabetic characters, digits, punctuation and some special characters)—today it also includes images, audios, videos and data from an exploding number of Internet of Things sensors in our homes, businesses, vehicles, cities and more.
- Veracity—the validity of the data—is it complete and accurate? Can we trust that data when making crucial decisions? Is it real?